

Jiarong (Bill) Lu

226-989-5986 · j94lu@uwaterloo.ca · <https://william-lu-jia-rong.github.io/>

RELEVANT SKILL

Languages: TypeScript, JavaScript, Python, Swift, C, C++, C#, SQL, HTML/CSS, Bash, VHDL

Frameworks & Libraries: Frontend: React, Next.js, React Native, SwiftUI, Three.js, Electron, Tailwind CSS; Backend: Node.js, Express, Fastify, FastAPI, Prisma, Drizzle ORM; AI/ML: OpenAI API, MCP servers, Ollama, Dify, OpenClaw, Apple Vision

Cloud & Data: PostgreSQL, Supabase (Auth, RLS, Realtime), SQLite, Cloudflare (Workers, R2), Vercel, Docker, Stripe, GitHub Actions

Tools: Cursor, Claude Code, Codex, Git/GitHub, VS Code, Xcode, Vitest, Playwright, STM32CubeIDE, Inventor, SolidWorks, Unreal Engine

EXPERIENCE

Studica Limited | Embedded Hardware & Firmware Developer

Jan 2026 - Apr 2026 | Sep 2026 - Dec 2026 (Expected)

- Built an AI-assisted robot configurator that imports company products directly into SolidWorks/Inventor or runs as a standalone web app, letting users assemble custom robots and order in one click
- Shipped as a React/Three.js web app, Electron desktop app, and C#.NET Inventor add-in, with 301 parts converted from STEP CAD files
- Built an AI-assisted visualization tool for CNC-based sensor calibration (Python, GRBL), replacing manual G-code workflows
- Developed and maintained firmware for five STM32-based products spanning motion, motor, distance, and color sensing
- Designed and laid out a compact 24 V, 10 A+ BLDC motor driver board (ø3.5 cm) running FOC

Automate Agency | Web & AI Automation Developer

Part-time | May 2026 - Present

- Designed and launched automateagency.ca, a responsive Next.js + Tailwind CSS website with an admin console, supporting service presentation and lead generation
- Built a daily Instagram workflow for post ideas, captions, creative assets, approval, scheduling, and publishing, connected with logs into an end-to-end content system
- Built an AI avatar video pipeline (TypeScript, HeyGen, ElevenLabs, FFmpeg/Remotion) that turns scripts into presenter videos

Glory Media | Software Developer

Remote | Jun 2025 - Aug 2025

- Built a native iOS social app for media subscribers with content sharing, messaging, events, and a marketplace, backed by Supabase (Postgres, Auth, Realtime) and Cloudflare R2
- Developed a Next.js + Supabase admin platform for managing users, posts, communities, events, and shop inventory
- Built a data platform aggregating influencer data from Instagram and LinkedIn with brand marketing data, applying AI models to generate marketing recommendations
- Designed OpenClaw skills and knowledge bases, enabling AI agents to function as domain-specific experts for enterprise deployment

Waterloo Aerial Robotics Group (WARG) | Embedded Flight Software Team Member

Oct 2024 - Oct 2026

- Built the motor driver for the new driver architecture and the Attitude Manager with PID control
- Modeled the thrust generated by the aircraft's motors and propellers

PROJECTS

RouteOS – LLM Model Router | TypeScript, Fastify, Supabase, Stripe, MCP

- Built a local-first router that predicts tokens, cost, and satisfaction across 38 models from 10 providers using a bundled MLP with SQLite feedback calibration, exposed via CLI, REST API, SDK, and MCP server
- Extended it into a hosted platform with an OpenAI-compatible auto-routing proxy, Supabase auth, a prepaid Stripe wallet, metered API keys, and AES-256-GCM encrypted provider credentials

PocketPilot – Expense Tracker for Apple Platforms | Swift, SwiftUI, FastAPI, Docker

- Built an offline-first expense tracker for iPhone, iPad, Mac, and Apple Watch that auto-captures Apple Pay/Wallet transactions via Shortcuts automations, on-device Vision OCR, and CSV import
- Added widgets, watch complications, duplicate detection, and optional sync to iCloud (CloudKit) or a self-hosted FastAPI + SQLite backend on a Raspberry Pi (Docker, Tailscale)

UW Course Planner | React, TypeScript, Vite, Tailwind CSS, pdf.js

- Built a web app that parses Quest transcript PDFs in the browser and checks prerequisites, corequisites, and antirequisites against Academic Calendar data to list eligible next-term courses
- Wrote a data pipeline from the calendar's Quali API and class schedules, plus career-track recommendations, degree-progress tracking, and UWFlow ratings

Agent-Native Game Engine | TypeScript, MCP, Electron, Three.js

- Built a TypeScript monorepo game engine whose MCP server lets Cursor, Codex, and Claude Code agents scaffold, patch, and verify levels, with deterministic 2D physics, a BFS solvability checker, and an Electron editor

EDUCATION

Candidate for Bachelor of Applied Science, University of Waterloo

Sept 2024 - Expected: June 2029

Bachelor of Applied Science in Computer Engineering